INTERNSHIP REPORT



Jodhpur Bus Stand Construction

Rajasthan State Road Development & Construction Corporation (RSRDC)

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SUBMITTED BY

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I express my sincere gratitude and appreciation for the opportunity to complete my internship at the Public Works Department (PWD) in Jodhpur. It has been an enriching and rewarding experience, and I am delighted to submit my internship report as a testament to the valuable knowledge and skills I have gained during my time at the organization.

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1 Introduction

Jodhpur, a city in the state of Rajasthan, India, is a vibrant and historic destination that captivates visitors with its rich cultural heritage and majestic architecture. Often referred to as the “Blue City” due to the prevalence of blue-painted houses in its old quarter, Jodhpur is a place where history, tradition, and modernity coalesce seamlessly.

Situated in the western part of Rajasthan, Jodhpur is the second-largest city in the state and serves as the administrative headquarters of the Jodhpur district. The city boasts a storied past that dates back centuries, having been founded in 1459 by Rao Jodha, the chief of the Rathore clan.

As you explore the streets, you’ll encounter a fascinating mix of old and new. The majestic Mehrangarh Fort, standing tall atop a hill, still commands respect and admiration, but it now shares the skyline with sleek high-rise buildings and modern structures that signify Jodhpur’s progress into the 21st century.

As a hub of education and technology, Jodhpur boasts top-notch universities and research institutions that attract students and professionals seeking knowledge and opportunities. The city’s commitment to education and innovation has nurtured a generation of young minds who are shaping its future.



Figure 1. Multi-storey building at the new Jodhpur bus terminal site

The need for an augmented bus terminal in Jodhpur arises from the city’s growing population, increasing tourist influx, and the demands of modern transportation infrastructure. While Jodhpur already has several bus terminals serving different routes, there are several reasons why a better transport/bus terminal would greatly benefit the city:

- **Tourism Growth:** Jodhpur is a popular tourist destination, attracting visitors from all over the world. A modern bus terminal can leave a positive first impression on tourists and enhance their overall experience. Tourists will appreciate well-designed waiting areas, clean facilities, and easy access to information.

- **Connectivity:** A better transport/bus terminal can enhance connectivity between Jodhpur and other cities and regions. Improved facilities and infrastructure can encourage more private and public bus operators to serve the city, leading to increased frequency and better route coverage.
- **Integration with Other Modes of Transport:** A well-planned terminal can integrate various modes of transport, such as buses, taxis, auto-rickshaws, and maybe even future public transportation projects like metro or light rail. This seamless integration would facilitate smoother transfers for passengers and encourage the use of public transport.
- **Economic Growth:** A well-planned and efficiently run bus terminal can have a positive impact on the local economy. It can attract more tourists, generate employment opportunities, and contribute to the overall development of the surrounding area.

2 Pavement: An Overview

A pavement for a bus terminal refers to the specially designed and constructed surface that provides a stable, durable, and safe platform for buses and other vehicles to maneuver, park, and load/unload passengers and cargo within the terminal premises. It serves as the primary area where buses and vehicles move and operate, making it a critical component of the overall infrastructure of a bus terminal.

The design and construction of the pavement for a bus terminal consider various factors to ensure its functionality and longevity. Some key aspects include:

1. **Load-Bearing Capacity:** The pavement must be capable of supporting the weight of buses and heavy vehicles without experiencing structural failure. The pavement's load-bearing capacity is determined by the type of materials used and the thickness of the pavement layers.
2. **Durability:** As a high-traffic area, the pavement must be durable enough to withstand constant use, including turning, braking, and the movement of heavy vehicles, without showing signs of premature wear or deterioration.
3. **Skid Resistance:** The pavement surface should offer sufficient skid resistance to prevent vehicles from slipping, especially during wet or slippery conditions, ensuring safe movement within the terminal area.
4. **Drainage:** Proper drainage is essential to prevent water accumulation on the pavement surface, which can lead to surface damage and safety hazards. Adequate drainage systems are designed to efficiently channel rainwater away from the pavement.
5. **Accessibility:** The pavement must be designed to accommodate the needs of all users, including pedestrians and individuals with disabilities, ensuring accessibility and safety for everyone.
6. **Markings and Signage:** Clear markings, signage, and lane demarcations are essential for guiding vehicles, pedestrians, and drivers, enhancing traffic flow, and minimizing congestion within the terminal area.
7. **Materials:** Various materials can be used for constructing the pavement, including concrete, asphalt, and interlocking pavers. The choice of materials depends on factors such as budget, climate, expected traffic volume, and maintenance requirements.



Figure 2. Prepared subgrade and formwork at the pavement area

8. **Maintenance:** Properly designed pavements should also be considered for ease of maintenance. Regular inspections and upkeep are crucial to prolonging the life of the pavement and ensuring the safety and functionality of the bus terminal.

A well-designed and constructed pavement for a bus terminal plays a vital role in ensuring smooth traffic flow, enhancing safety, and providing a pleasant experience for passengers, drivers, and visitors using the terminal facilities.

3 Procedure to Construct a Pavement

Constructing a cement concrete pavement for a bus terminal involves several systematic steps to ensure a durable and well-constructed surface capable of withstanding heavy vehicle loads and providing a smooth and safe platform for buses and other vehicles. Below is a step-by-step procedure for constructing a cement concrete pavement for a bus terminal:

1. Site Preparation

- Conduct a thorough survey and analysis of the site to determine the layout, dimensions, and required thickness of the pavement.
- Clear the site of any vegetation, debris, or obstacles that might hinder construction.
- Establish the desired levels and slopes for proper drainage.

2. Excavation and Subgrade Preparation

- Excavate the area to the required depth to establish the subgrade, ensuring it is stable and able to support the pavement.
- Compact the subgrade thoroughly to achieve the necessary load-bearing capacity.
- Rectify any soft spots or weak areas by adding suitable materials and compacting them.

3. Formwork Installation

- Erect formwork around the edges of the pavement area to confine the concrete during pouring.
- The formwork should be properly aligned, level, and securely fixed to prevent the concrete from spilling or deforming during the pouring process.
- A plastic sheet of 125 microns is also laid to prevent water through ground seepage to affect the strength of pavement.



Figure 3. Formwork, reinforcement and plastic sheet laid before concreting

4. Reinforcement

- Dowel bars are short steel bars that are used in concrete pavements and other concrete structures to transfer loads between adjacent concrete slabs. They play a vital role in providing load transfer and preventing the formation of cracks at the joints, which can occur due to temperature variations, traffic loads, and other factors. Dowel bars accommodate thermal expansion and contraction of the concrete slabs, allowing them to move without excessive stresses that could lead to cracking. They are placed every 3–4.5-meter length of pavement as per IS code.



Figure 4. Dowel bars placed at the joint over the plastic sheet

- Tie bars, also known as transverse reinforcement or crossties, are steel bars used in reinforced concrete structures to resist tension forces and prevent the development of cracks. They play a crucial role in enhancing the structural integrity, durability, and load-carrying capacity of concrete elements.

5. Concrete Mix Design

- Design a suitable concrete mix that meets the required strength, durability, and workability for the pavement.
- Factors such as the local climate, expected traffic loads, and availability of materials should be considered in the mix design.

6. Concrete Pouring

- Pour the concrete mix into the formwork systematically, ensuring even distribution and avoiding voids.
- Compact the concrete thoroughly using vibrating equipment to remove air bubbles and achieve maximum density.



Figure 5. Concrete pouring and compaction within the formwork

- Floaters are used to smoothen the surface and dewatering is also performed through vacuum.

7. Finishing



Figure 6. Broom-finished concrete surface with grooves

- Use straightedges and screeds to level the surface of the freshly poured concrete, ensuring a uniform and smooth finish.
- Allow the concrete to partially set before applying a broom finish or other texturing methods to improve skid resistance. Grooves of 2 mm.

8. Curing

- Proper curing is crucial for the strength and durability of the pavement. Apply a curing compound or cover the surface with wet burlap or a curing membrane to retain moisture and prevent premature drying.

9. Expansion Joints



Figure 7. Expansion joint cut in the finished concrete pavement

- Install expansion joints at appropriate intervals to allow for thermal expansion and contraction of the concrete pavement.
- Expansion joints are to be made within 24 hours of finishing and they are typically 2–3 mm.

By following this systematic procedure, the construction of a cement concrete pavement for a bus terminal can be carried out efficiently, resulting in a durable and reliable surface that may seem costlier due to initial cost, but maintenance is inversely proportional to the cost.

4 Expansion Joint



Figure 8. Expansion joint between adjacent building components

An expansion joint in a building is a deliberate separation or gap created between adjacent building components to accommodate the movement caused by thermal expansion, contraction, and other factors. These joints play a critical role in preventing structural damage and maintaining the integrity of the building over time. Expansion joints are particularly important in large and tall structures where thermal movement can be significant.

Key features and functions of expansion joints in buildings include:

1. **Movement Accommodation:** Buildings are subject to various forms of movement, such as thermal expansion and contraction due to temperature fluctuations, seismic activity, settlement, and creep. Expansion joints are designed to allow controlled movement of building components without causing stress, cracking, or deformation.
2. **Types of Expansion Joints:** There are several types of expansion joints used in buildings, including:
    a. *Horizontal Expansion Joints:* These are placed at intervals along the length or width of the building to allow for horizontal movement.
    b. *Vertical Expansion Joints:* These are used to accommodate vertical movement between different parts of the building, especially in tall structures. Typically, between every 25–30 m stretch.
    c. *Seismic Expansion Joints:* Designed to absorb and distribute seismic forces, these joints play a crucial role in earthquake-resistant building design.

3. **Materials and Design:** Expansion joints are typically constructed using flexible materials, such as rubber, neoprene, or metal, that can absorb and accommodate movement. EPDM sheets (Ethylene Propylene Diene Monomer) are installed to prevent flow of water from the precipitation and is bonded together by polysulphide (resins).



Figure 9. Close-up of the expansion joint gap with EPDM sheet and sealant

4. **Location:** Expansion joints are strategically placed in areas where movement is likely to occur, such as between different building segments, floors, and structural elements like walls and slabs.
5. **Aesthetics:** While functional, expansion joints can also have aesthetic considerations. Architects and designers often integrate these joints into the building's overall design to maintain a pleasing visual appearance.
6. **Maintenance:** Proper maintenance of expansion joints is essential to ensure their effectiveness. Regular inspections and repairs, if necessary, help prevent water intrusion, debris accumulation, and deterioration of joint materials.
7. **Fire Protection:** In certain cases, expansion joints might need to provide fire-resistant properties, particularly in buildings where fire compartmentalization is required for safety reasons.

In summary, expansion joints in buildings serve a critical purpose by accommodating movement caused by thermal expansion, contraction, seismic activity, and other factors. Properly cracking and installed expansion joints help prevent structural damage, minimize cracking, and extend the lifespan of the building as it could even affect even the foundation. They are an integral part of building construction, ensuring the safety, stability, and longevity of the structure while providing the necessary flexibility to accommodate the forces at play.

5 Utilities at the New Bus Terminal

These utilities cater to the diverse needs of the terminal, including transportation services, passenger amenities, safety, and overall functionality. Some of the key utilities at a bus terminal in Jodhpur may include:

1. **Bus Bays and Stands:** Designated bus bays and stands where buses can park and load/unload passengers. Properly marked and organized bays help streamline traffic flow within the terminal.
2. **Ticket Counters:** Ticketing facilities where passengers can purchase bus tickets and obtain information on schedules, fares, and available routes. In total 24 counters are proposed and one additional counter for advance booking is also there.



Figure 10. Interior of the terminal hall where the ticket counters are proposed

3. **Waiting Areas:** Fully air-conditioned and comfortable waiting areas with seating arrangements for passengers to rest and wait for their buses.
4. **Restrooms:** Clean and well-maintained restrooms for both men and women, offering essential facilities for passengers.
5. **Police Station:** Separate police stations for male and female to maintain law and order, and also provide immediate grievance redressal.
6. **Cafeteria/Food Stalls/Hotels:** Eating areas or food stalls offering a variety of refreshments and snacks for passengers and visitors.
7. **Parking Area:** Adequate parking space for private vehicles and two-wheelers belonging to passengers and staff.
8. **Passenger Information Display Systems:** Digital display boards or screens showing bus schedules, routes, and other essential announcements.
9. **Security Measures:** Surveillance cameras and security personnel to ensure the safety and security of passengers and their belongings.

10. **Drinking Water Facilities:** Clean drinking water dispensers or water fountains for passengers' convenience.
11. **Fire Safety Measures:** Fire safety equipment, including fire extinguishers and fire alarm systems, to ensure safety in case of emergencies.
12. **Accessibility Features:** Facilities to ensure accessibility for passengers with disabilities, such as ramps, handrails, and designated parking spaces.
13. **Maintenance and Workshop Facilities:** Facilities for bus maintenance and repair, including workshop areas and service bays.

These utilities collectively contribute to creating a well-organized, passenger-friendly, and efficient bus terminal in Jodhpur, catering to the needs of travelers and promoting a positive travel experience. Properly managed utilities enhance the terminal's reputation and contribute to the overall development of the city's transportation infrastructure.

6 Conclusion & Discussions

A very enriching experience to observe minute little things involved in the process of going through on-field training. It is fortunate enough to be part of a project which would directly deal with and will be of use to public, understanding the needs helps broaden the horizons and make us capable to think in broad terms.

The proposed 4-foot green corridor around the boundary of bus stand premises not only has aesthetic benefit but it echoes the need for sustainable development and showcases itself to promote green spaces.

There is also provision of multi-level parking where the old bus stand operates, which is the best fit for the growing future demands of the city and rules out the congestion. It also sets an example for proper land use of existing infrastructure and apt utilization of resources.

Also, as the number of electric vehicles will burgeon the existing fleets, there ought to be provision for electric charging ports.

The new bus terminal will help the growing city of Jodhpur transition towards a multi-modal transportation system, which would help reduce carbon emissions.



Figure 11. Illustration of an integrated, multi-modal transportation system